15 Prove algebraically that the sum of the squares of any two consecutive even numbers is always a multiple of 4

Let consecutive even numbers be $2n$ and $2n+2$.

Their squared sum is $(2n)^2 + (2n+2)^2$

$= 4n^2 + 4n^2 + 8n + 4$

$= 4(2n^2 + 2n + 1)$, a multiple of 4.

3 marks total
Represent both evens
Expand both squares
Factor out 4

(Total for Question 15 is 3 marks)

16 y is inversely proportional to the square of x.

$y = 8$ when $x = 2.5$

Find the negative value of x when $y = \frac{8}{9}$

.....

(Total for Question 16 is 3 marks)

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